MATH 340

Quiz 3

February 14, 2025

1. (10 points) Problem 1

   (a) Find the general solution to the homogeneous ODE $x'' - 4x' + 13x = 0$.

   (b) Find the general solution to the homogeneous ODE $x'' - 4x' + 3x = 0$.

   (c) Find a particular solution of the nonhomogeneous ODE $x'' - 4x' + 3x = e^t$.

2. (10 points) Problem 2. Consider the 2nd order ODE $x'' + 4x = \sin(t)$.

   (a) Does the ODE model a free oscillation? Is the oscillation over-damped? (We will do this one in class!)

   (b) Find a particular solution of the above nonhomogeneous ODE.

   (c) Find the solution of the ODE that satisfies the initial conditions $x(0) = 1, x'(0) = 0$.